

Talk 01 — Homework

Course Overview & AI-assisted Setup

Your Name — Your Student ID

2026-09-03

Table of contents

Instructions 作业说明	1
User Verification 用户身份验证	2
0.1 Talk 01 Content Review 内容回顾	2
Exercise 1: System Verification	3
0.1 1.1 Check your R version	3
0.2 1.2 Check your operating system	3
0.3 1.3 Check your working directory	3
0.4 1.4 List installed packages	4
0.5 1.5 Verify key packages	4
0.6 1.6 Verify tinytex	4
Exercise 2: AI-assisted Workflow Practice	5
0.1 2.1 Reproduce the verification-beamer PDF	5
0.2 2.2 Create a homework-style verification PDF	5
Exercise 3: AI Prompt Practice	6
0.1 3.1 Writing effective prompts	6
0.2 3.2 Reflect on your AI experience	7
Exercise 4: Prompt Cheat Sheet	7

Instructions 作业说明

- Write your R code inside `{r}` code chunks, run them, and verify the output. 将 R 代码填入 `{r}` 代码块中，运行并验证输出结果。

- After completing all exercises, click the **Render** button in RStudio (or run `quarto render`) to generate a PDF. 完成后, 点击 RStudio 中的 **Render** 按钮生成 PDF。
- **Rename the PDF** to: `Name-StudentID-talk01-homework.pdf` and submit to the course platform. 将 **PDF** 重命名为: 姓名-学号-talk01-homework.pdf, 提交到课程平台。
- **Important:** If your system username does not reflect your real name, change it to your Pinyin name before running this homework. 注意: 如果你的系统用户名不是真实姓名拼音, 请在运行作业前修改。

User Verification 用户身份验证

Run the following to verify your identity 运行以下代码验证你的身份:

```
Sys.info()[["user"]]
```

```
[1] "weihuachen"
```

```
Sys.getenv("HOME")
```

```
[1] "/Users/weihuachen"
```

0.1 Talk 01 Content Review 内容回顾

Today we covered 今天的内容包括:

- Course overview and structure (16 talks, two parts) 课程概述与结构 (16 讲, 两部分)
- Why R? — history, popularity, bioinformatics advantages 为什么选 R? ——历史、流行度及生物信息学优势
- AI-assisted programming philosophy with **CodeBuddy** AI 辅助编程理念 (使用 **CodeBuddy**)
- Step-by-step setup: CodeBuddy → R → RStudio → Extensions → tinytex → R packages → tidyverse 逐步安装流程
- Clone the course repository and open the R Project 克隆课程仓库并打开 R Project
- Render QMD files to PDF (Beamer & Article formats) 渲染 QMD 文件为 PDF (Beamer 与 Article 格式)

Exercise 1: System Verification

0.1 1.1 Check your R version

Print your R version string using `R.version.string`:

```
## Write your code here and run it  
R.version.string
```

```
[1] "R version 4.5.0 (2025-04-11)"
```

0.2 1.2 Check your operating system

Print your operating system details using `Sys.info()`:

```
## Write your code here and run it  
Sys.info()
```

```
"Darwin Kernel Version 25.6.0: Fri Jul 31 19:11:03 PDT 2026; root:xnu-12377.161.14~5/RELEASE_ARM_T8040"
```

```
"Weihuas"
```

```
e:
```

0.3 1.3 Check your working directory

Print your current working directory with `getwd()`. Does it point to the course repository?

```
## Write your code here and run it
```

Question: What is your current working directory? Is it the cloned course repository (~/Desktop/R-for-bioinformatics)? 你当前的工作目录是什么? 是否为克隆的课程仓库?

Answer:

```
;lj;j;l
```

0.4 1.4 List installed packages

Create a data frame (using `tibble`) that lists **all** installed R packages, with columns: `Package`, `Version`, `LibPath`. Show the first 20 rows.

```
## Write your code here and run it
```

0.5 1.5 Verify key packages

Check whether the following packages are installed and report their versions:

- `rmarkdown`
- `knitr`
- `tidyverse`
- `tinytex`

```
## Write your code here and run it
```

Question: Are all four packages installed? If any is missing, use CodeBuddy to install it and report the version after installation.

Answer:

0.6 1.6 Verify tinytex

Check whether TinyTeX is properly installed:

```
## Write your code here and run it
```

Question: Does `tinytex::is_tinytex()` return `TRUE`? If not, ask CodeBuddy to help you install it.

Answer:

Exercise 2: AI-assisted Workflow Practice

0.1 2.1 Reproduce the verification-beamer PDF

Use **CodeBuddy** to help you create a new QMD file (save it as `my-verification-beamer.qmd` in your project root). The file should:

- Use **Beamer** format (1610 aspectratio, Warsaw theme, xelatex engine)
- Contain a title slide: “My Verification Report — Beamer”
- Show your system info (user name, R version, platform)
- Show your working directory
- List at least 10 installed packages

After CodeBuddy creates the file, **render it in RStudio** and confirm the PDF is generated correctly.

```
# No R code needed — this is a QMD creation + rendering exercise
# Paste the CodeBuddy prompt you used below:
```

CodeBuddy prompt used (paste here):

Question: Did the PDF render successfully? Describe any issues you encountered and how you resolved them (with CodeBuddy’s help or on your own).

Answer:

0.2 2.2 Create a homework-style verification PDF

Use **CodeBuddy** to help you create another QMD file (save as `my-verification-homework.qmd`). This time, use **article format** (`format: pdf`). The file should:

- Have a top-level heading: “My Verification Report — Homework Style”
- Contain sections showing: user name, R version, OS, working directory
- Include a table of at least 10 installed packages (Package, Version, LibPath)
- Have `echo: true` on all code chunks

After CodeBuddy creates the file, **render it in RStudio**.

```
# No R code needed — paste your CodeBuddy prompt below:
```

CodeBuddy prompt used (paste here):

Question: Describe the key differences between rendering a Beamer QMD vs. an Article QMD. Which format would you prefer for homework submissions?

Answer:

Exercise 3: AI Prompt Practice

0.1 3.1 Writing effective prompts

For each scenario below, write an **effective CodeBuddy prompt** that would accomplish the task. Then (optionally) try it in CodeBuddy and report the result.

Scenario A: You want to install a new R package called `palmerpenguins`.

Your prompt:

Scenario B: You get an error message when trying to render a QMD file: `"Quitting from lines 12-15 (myfile.qmd) Error in library(ggplot2): there is no package called 'ggplot2'".` Write a prompt to debug this.

Your prompt:

Scenario C: You want CodeBuddy to explain the differences between R Markdown (`.Rmd`) and Quarto (`.qmd`) files.

Your prompt:

0.2 3.2 Reflect on your AI experience

Question: In this homework, which tasks did you use CodeBuddy for? What worked well? What was challenging? 本次作业中你在哪些任务中使用了 CodeBuddy? 哪些效果好? 哪些有挑战?

Answer:

Exercise 4: Prompt Cheat Sheet

Based on your experience so far, create your personal **prompt cheat sheet** with at least 5 prompt templates you find useful. Refer to the lecture slides for inspiration.

```
# No code needed — write your cheat sheet below:
```

My Prompt Cheat Sheet:

Scenario	My Prompt Template
Install package	
Debug error	
Explain concept	
Create file	
Run analysis	

Homework ends here. Double-check that all code chunks run without errors, then Render to PDF. 作业结束。请确认所有代码块运行无误, 然后 Render 生成 PDF。